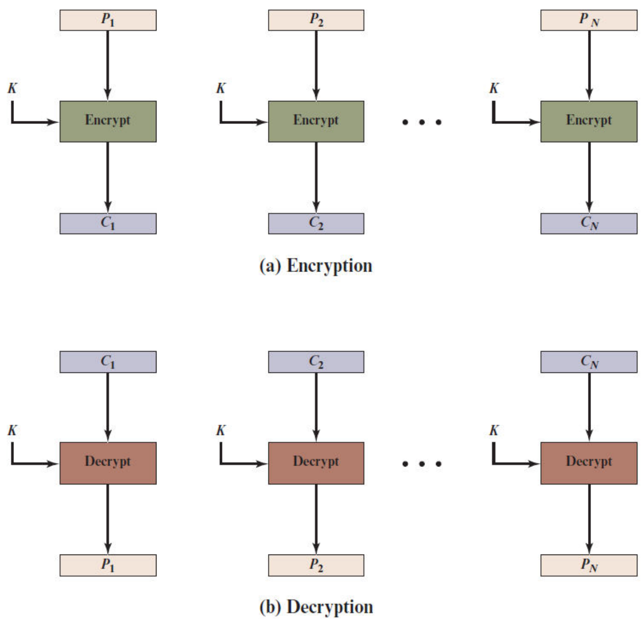
Block Cipher Operations

- There are a number of different ways to apply a block cipher to plaintext.
    - Electronic Codebook (ECB)
    - Cipher Block Chaining (CBC)
    - Cipher Feedback (CFB)
    - Output Feedback (OFB)
    - Counter (CTR)
- Mode of operation is a technique for enhancing the effect of a cryptographic algorithm or adapting the algorithm for an application, such as applying a block cipher to a sequence of data blocks or a data stream.

1. Electronic Codebook

- Plaintext is **handled one block at a time** and each block of plaintext is encrypted using the same key.
- The term codebook is used because, for a given key, **there is a unique ciphertext for every b-bit block of plaintext.**
- For a **message longer than b bits, the procedure is simply to break the message into b-bit blocks**, padding the last block if necessary.
- Decryption is performed one block at a time, always using the same key.



ECB	$C_j = E(K, P_j)$	$j = 1, \ldots, N$	$P_j = D(K, C_j)$	$j = 1, \ldots, N$

1. Electronic Codebook (Contd…)

- **Significant Characteristic of ECB**
  - Same **b-bit block of plaintext appears more than once** in the message, it **always produces the same ciphertext**.
- The ECB mode should be used only to **secure messages shorter than a single block** of underlying cipher.
  - Because in most of the cases messages are longer than the encryption block mode, this mode has a minimum practical value.
  - **For lengthy messages, the ECB mode may not be secure.**
- If **the message is highly structured**, it may be possible for a **cryptanalyst to exploit these regularities**.
  - Starting and Ending with the same text.
  - Same text repeating multiple times in the plain text.

Advantages of using ECB

- **Parallel encryption of blocks of bits is possible**, thus it is a faster way of encryption.
- Simple way of the block cipher.

Disadvantages of using ECB

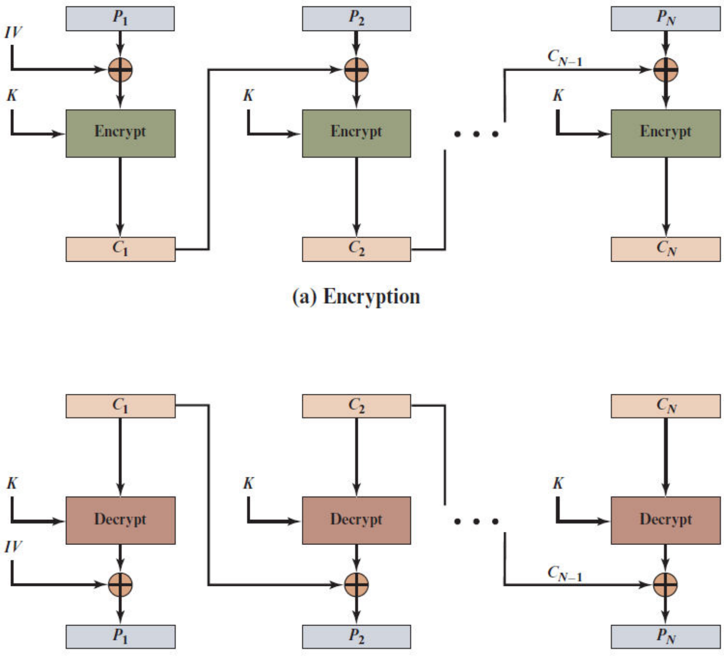
- Prone to **cryptanalysis** since there is a direct relationship between plaintext and ciphertext.

Properties for evaluating and constructing block cipher modes

- These properties are **applied to modes superior to ECB.**
- **Overhead:**
  - The **additional operations for the encryption and decryption** operation when compared to encrypting and decrypting in the ECB mode.
- **Error recovery:**
  - The property that **an error in the ith ciphertext block is inherited by only a few plaintext blocks** after which the mode resynchronizes.
- **Error propagation:**
  - The property that an **error in the ith ciphertext block is inherited by the ith and all subsequent plaintext blocks**.
- **Diffusion:**
  - How the **plaintext statistics are reflected in the ciphertext**.
- **Security:**
  - Whether or not the **ciphertext blocks leak information about the plaintext blocks**.

2. Cipher Block Chaining Mode (CBC)

- To overcome the security deficiencies of ECB, **we would like a technique in which the same plaintext block, if repeated, produces different ciphertext blocks**.
- **Input** to the encryption algorithm is the **XOR of the current plaintext** block and the **preceding ciphertext block.**
- The **same key is used for each block**.
- **Therefore, repeating patterns of b bits are not exposed.**
- For **decryption**, each cipher block is passed through the decryption algorithm.
- The **result is XORed with the preceding ciphertext block** to produce the plaintext block.



(a) Encryption

(b) Decryption

2. Cipher Block Chaining Mode (CBC) (Contd…)

CBC	$C_1 = E(K, [P_1 \oplus IV])$ $C_j = E(K, [P_j \oplus C_{j-1}])j = 2, \ldots, N$	$P_1 = D(K, C_1) \oplus IV$ $P_j = D(K, C_j) \oplus C_{j-1} j = 2, \ldots, N$

- The **IV (Initialization Vector) is a data block that is the same size as the cipher block**.
- The **IV must be known to both the sender and receiver** but be **unpredictable by a third party.**
- For maximum security, the IV should be protected against unauthorized changes.
- This could be done by sending the IV using ECB encryption.

Advantages of CBC

- CBC **works well for input greater than b bits.**
- CBC is a good authentication mechanism.
- **Better resistive nature towards cryptanalysis** than ECB.

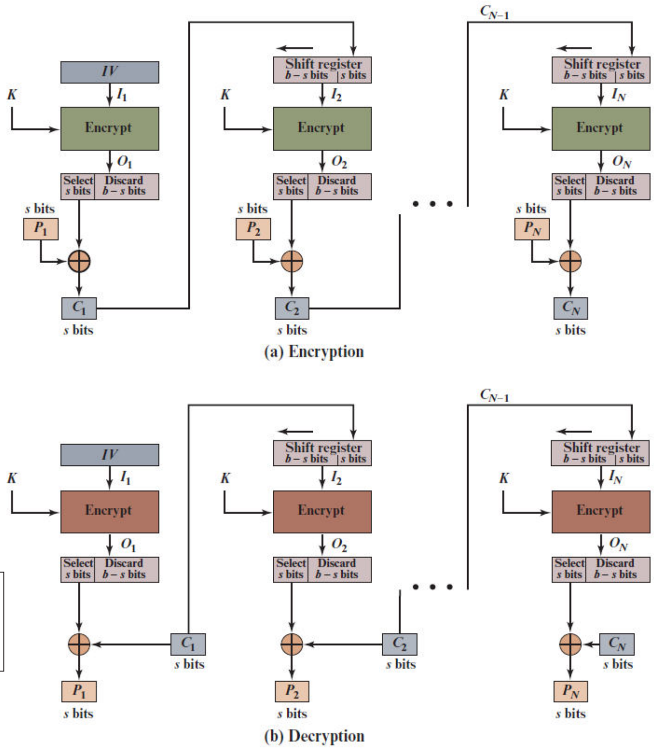
Disadvantages of CBC

- **Parallel encryption is not possible** since every encryption requires a previous cipher.

3. Cipher Feedback Mode (CFB)

- This mode is used to convert a block cipher into stream cipher.
- Rather than blocks of b bits, the **plaintext is divided into segments of s bits**.
- We get segments of Cipher Text of length s bits.

CFB	$I_1 = IV$ $I_j = \text{LSB}_{b-s}(I_{j-1}) \parallel C_{j-1} \quad j = 2, \ldots, N$ $O_j = \text{E}(K, I_j) \quad j = 1, \ldots, N$ $C_j = P_j \oplus \text{MSB}_s(O_j) \quad j = 1, \ldots, N$	$I_1 = IV$ $I_j = \text{LSB}_{b-s}(I_{j-1}) \parallel C_{j-1} \quad j = 2, \ldots, N$ $O_j = \text{E}(K, I_j) \quad j = 1, \ldots, N$ $P_j = C_j \oplus \text{MSB}_s(O_j) \quad j = 1, \ldots, N$



Advantages of CFB

- Since, there is some data loss due to the use of shift register, thus it is **difficult for applying cryptanalysis**.

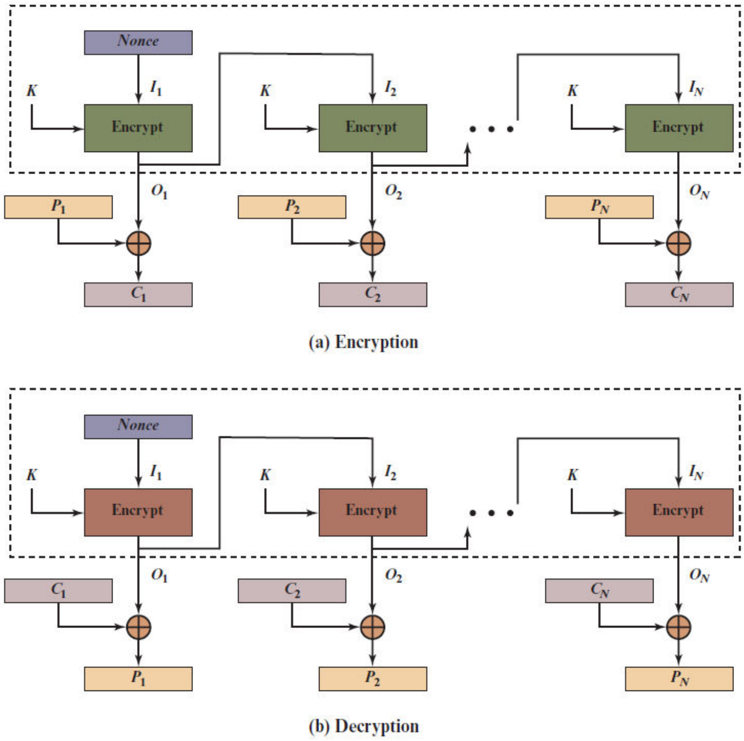
Disadvantages of CFB

- The drawbacks of CFB are the same as those of CBC mode.
- Both **block losses and concurrent encryption of several blocks are not supported** by the encryption.
- Decryption, however, is parallelizable and loss-tolerant.

4. Output Feedback Mode (OFB)

- Operation is similar to that of CFB mode.
- Nonce value is already known.

OFB	$I_1 = Nonce$ $I_j = O_{j-1} \quad j = 2, \ldots, N$ $O_j = E(K, I_j) \quad j = 1, \ldots, N$ $C_j = P_j \oplus O_j \quad j = 1, \ldots, N-1$ $C_N^* = P_N^* \oplus MSB_u(O_N)$	$I_1 = Nonce$ $I_j = O_{j-1} \quad j = 2, \ldots, N$ $O_j = E(K, I_j) \quad j = 1, \ldots, N$ $P_j = C_j \oplus O_j \quad j = 1, \ldots, N-1$ $P_N^* = C_N^* \oplus MSB_u(O_N)$



Advantages of OFB

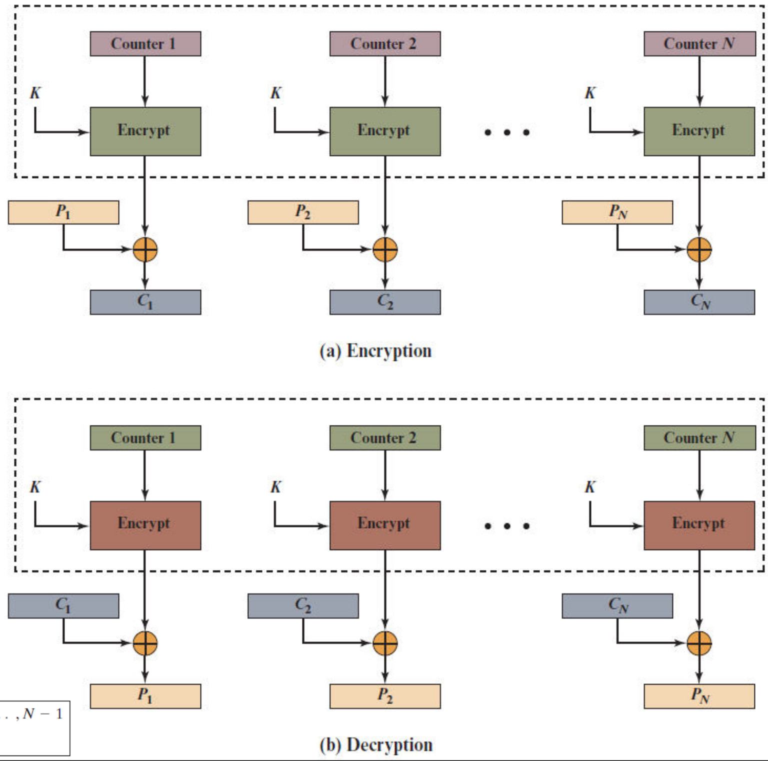
- In the case of CFB, a single bit error in a block is propagated to all subsequent blocks.
- This problem is solved by **OFB as it is free from bit errors in the plaintext block.**

Disadvantages of OFB

- The drawback of OFB is that, because to its operational modes, it is more susceptible to a **message stream modification attack** than CFB.

5. Counter Mode (CTR)

- **A counter equal to the plaintext block size is used.**
- The **counter value must be different for each plaintext** block that is encrypted.
- Typically, the counter is **initialized to some value and then incremented by 1** for each subsequent block (modulo 2^b, where b is the block size).



(a) Encryption

(b) Decryption

CTR	$C_j = P_j \oplus E(K, T_j) \quad j = 1, \ldots, N-1$ $C_N^* = P_N^* \oplus MSB_u[E(K, T_N)]$	$P_j = C_j \oplus E(K, T_j) \quad j = 1, \ldots, N-1$ $P_N^* = C_N^* \oplus MSB_u[E(K, T_N)]$

Advantages of CTR

- **Hardware efficiency**
  - Parallel processing is possible.
- **Software efficiency**
  - As parallelization is possible, aggressive pipelining takes place internally which reduces the time for completion.
- **Preprocessing**
  - The execution of the underlying encryption algorithm does not depend on input of the plaintext or ciphertext.
- **Random access**
  - The ith block of plaintext or ciphertext can be processed in random-access fashion.
- **Simplicity**
  - No decryption algorithm is used.

Disadvantages of CTR

- The fact that CTR mode requires a synchronous counter at both the transmitter and the receiver is a severe drawback.
- The recovery of plaintext is erroneous when synchronization is lost.

Comparison of Block Cipher Operations

Table 7.1 Block Cipher Modes of Operation

Mode	Description	Typical Application
Electronic Codebook (ECB)	Each block of plaintext bits is encoded independently using the same key.	• Secure transmission of single values (e.g., an encryption key)
Cipher Block Chaining (CBC)	The input to the encryption algorithm is the XOR of the next block of plaintext and the preceding block of ciphertext.	• General-purpose block-oriented transmission • Authentication
Cipher Feedback (CFB)	Input is processed *s* bits at a time. Preceding ciphertext is used as input to the encryption algorithm to produce pseudorandom output, which is XORed with plaintext to produce next unit of ciphertext.	• General-purpose stream-oriented transmission • Authentication
Output Feedback (OFB)	Similar to CFB, except that the input to the encryption algorithm is the preceding encryption output, and full blocks are used.	• Stream-oriented transmission over noisy channel (e.g., satellite communication)
Counter (CTR)	Each block of plaintext is XORed with an encrypted counter. The counter is incremented for each subsequent block.	• General-purpose block-oriented transmission • Useful for high-speed requirements